Real-Time Blood Pressure Prediction Using Apache Spark and Kafka Machine Learning

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Abstract— Using a mix of machine learning algorithms and big data tools, particularly Apache Spark and also Apache Kafka, this research provides a new method for real-time blood pressure prediction. The method can handle large amounts of inbound data from numerous sources, including wearable technology and internet of things monitors. A clustering-based approach is used to improve the blood pressure estimation's precision while the data is being analyzed in real-time. ECG, PPG, and ABP signals dataset are used to assess the suggested strategy, and the findings show a substantial improvement in blood pressure prediction accuracy when compared to previous methods. The suggested method has the potential to be used in numerous uses, such as remote patient tracking, individualized healthcare, and cardiovascular disease early detection. This research offers two contributions. First off, it introduces a novel technique for real-time blood pressure forecast that is more accurate than current approaches. In addition, it shows the value of merging machine learning techniques with real-time streaming data processing systems like Apache Spark and Apache Kafka. Further improving the scalability and accuracy of the system is the use of web-based tools and deep learning methods. The suggested method may have a big impact on how well patients do and how much it will cost to treat them. Overall, this research offers a path that can be useful to both individuals and healthcare professionals for the creation of real-time blood pressure forecast tools.

Keywords— Real-time prediction, blood pressure, machine learning, streaming data, wearable devices, IoT sensors, clustering, web-based platform

I. INTRODUCTION

More than one billion individuals globally are affected by hypertension, also referred to as high blood pressure[1]. Renal failure, heart disease, and stroke are all significantly increased by uncontrolled hypertension[2, 3]. Therefore, precise and timely blood pressure tracking is essential for the effective control of illness and the avoidance of related complications. Recent advancements in wearable technology and Internet of Things monitors have opened up new possibilities for constant, non-invasive blood pressure monitoring[4]. The precision of real-time blood pressure forecast has increased as a result of the merging of these devices with machine learning algorithms. Apache Spark, a distributed computing engine that offers quick and effective processing of streaming data, is one of the most promising technologies for managing huge volumes of real-time data [5, 6]. To manage the large amount of incoming data from different sources, Apache Spark is frequently used in combination with Apache Kafka, a communications system that is decentralized and has a high throughput. Numerous studies have been conducted to investigate the feasibility of using machine learning algorithms to make predictions about blood pressure. Several research, for instance, have shown how machine learning algorithms can increase the precision of blood pressure prediction. Furthermore, it has been demonstrated that the accuracy of healthcare estimation is improved by the combination of clustering methods and machine learning models. For instance, utilizing ECG and PPG signals, clustering-based technique was suggested for blood pressure prediction[7]. The result showed that, in comparison to earlier techniques, this approach greatly increased the accuracy of blood pressure prediction. Many possible uses for the use of real-time ML algorithms for blood pressure prediction include remote patient monitoring, individualized healthcare, and the early detection of cardiovascular illnesses. Healthcare workers may be able to monitor patients' blood pressure constantly and be alerted to any aberrant readings thanks to the development of a web-based platform for real-time blood pressure monitoring and analysis[8]. This would improve patient outcomes and save healthcare expenditures. Can a combination of machine learning algorithms and streaming data processing tools, specifically Spark and Kafka, be used to better estimate blood pressure in real time? In the current work, we describe a unique method for blood pressure estimation in real-time that combines machine learning algorithms with platforms for streaming data, such as Spark and Kafka. To enhance the precision of BP estimate, we employ clustering-based techniques. We test our method using a dataset of ECG and PPG signals. In comparison to earlier methods, our results show a considerable increase in the accuracy of blood pressure prediction. Additionally, we develop platform for real-time blood pressure monitoring to create a notification for alerting users to any abnormal readings. The reason for choosing the proposed combined method is the need to handle large amounts of inbound data from numerous sources, including wearable technology, healthcare devices and Internet of Things monitors, in real-time. The precision of blood pressure prediction can be improved by employing clustering-based techniques and the using of ML algorithms. Furthermore, the usage of large and streaming data processing tools like Apache Spark and Apache Kafka offers a promising technology for managing huge volumes of real-time data. The suggested method has the potential to be used in numerous uses, such as remote patient tracking, individualized healthcare, and cardiovascular disease early detection. The following describes the structure of the paper: Section II presents a comprehensive review of the literature on Blood Pressure Measurement. The methodology of our prediction model is described in Section III. Section IV presents the experimental outcomes derived from the application of our model. Section V concludes the study by reaching a conclusion based on the findings.

II. RELATED WORKS

Some research have looked towards real-time blood pressure prediction with machine learning algorithms in recent years. We discuss some of the most important works in this field in this section. A real-time health care surveillance system that utilizes Apache Kafka and Apache Spark, was suggested by Hassan et al.[9]. The system collects data from multiple sources, for instance: wearable technology and IoT sensors, and uses Apache

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Spark to analyse the data in real time. To forecast heart rate, blood pressure, and other variables, the authors utilized a regression model. The outcomes showed that their method had a high degree of predictive accuracy. Nandi et al. [10] conducted another investigation. examined the application of deep learning techniques to prediction of BP. Using an ECG and PPG dataset, the authors constructed a deep learning model utilizing CNNs and LSTM networks. The findings demonstrated that their model had a significant achievement in SBP and DBP estimation. A Deep Learning-based blood pressure prediction system that made use of a combination of PPG signals was created by Zhao et al. [11] in a different study. The authors predicted blood pressure using convolutional neural networks (CNN, CNN-GRU, and CNN-LSTM). The outcomes showed that their strategy had a high degree of accuracy in forecasting blood pressure.

The application of a CNN networks for blood pressure estimation was examined in another research by Cheng et al.[12]. The authors created a 1-D CNN model using a PPG dataset. According to the findings, their model was highly accurate at predicting both systolic and diastolic blood pressure. Also, in certain studies[13, 14], the authors offered numerous unique methods for predicting blood pressure that combined ECG and PPG data with newly derived variables including HR and PIR. The outcomes showed that their method was highly accurate at predicting blood pressure. Apache Spark was used in two of the aforementioned research [15, 16] to process real-time data and forecast blood pressure, illustrating the value of fusing machine learning algorithms with streaming data processing platforms. In conclusion, the experiments examined in the current section show the capability of streaming data processing platforms like Apache Spark and ML algorithms to predict real-time blood pressure. The suggested methods make use of multiple data types, such as ECG and PPG, and diverse ML algorithms, including deep learning and clustering-based algorithms. The findings demonstrate that these methods can predict blood pressure with a high degree of accuracy and suggest that they may have important implications for bettering healthcare outcomes.

III. PROPOSED WORK

This research proposes a new combination method for predicting real-time blood pressure using a combination of machine learning algorithms and streaming data processing platforms. The proposed approach will utilize Apache Spark and Apache Kafka to handle the high volume of incoming signals from various sources, including wearable technology and IoT sensors. The data will process in real-time, and a clustering-based algorithm will use to improve blood pressure estimation precision. The proposed system's architecture is depicted in Fig.1.

Phase 1: Machine Learning Model Development using Apache Spark:

1) *Load the MIMIC II dataset into a Spark DataFrame.*

2) Extract features of independent and dependent variables using Spark's built-in functions and libraries.

3) Split the features into train and test categories using Spark's randomSplit() method.

4) Dretermine the best number of clusters by clustering the extracted features of PTT, PIR, and HR in train using the K-means algorithm and the silhouette criterion.

5) Using Spark's MLlib library, develop RFR ML model over each of the clusters.

6) Evaluate each cluster's model using Spark's built-in evaluation metrics.

7) Evaluate models for the test category using Spark's built-in evaluation metrics.

Phase 2: Real-time Data Processing and Web-based Platform Development using Apache Spark:

1) *Collect hospital data from various sources, including wearable gadgets and IoT sensors.*

2) Stream preprocessed data using Apache Kafka.

3) Preprocess the received signals and remove noise and artifacts using Spark's built-in signal processing libraries.

4) Process data in Apache Spark and extract features such as HR, PIR, and PTT using Spark's built-in functions and libraries.

5) Use models developed in the first phase to estimate SBP and DBP from extracted features using Spark's MLlib library.

6) Evaluate machine learning models' performance using metrics such as rmse and mae using Spark's built-in evaluation metrics.

7) Update machine learning models for each cluster in real-time as new data becomes available using Spark's streaming capabilities.

8) Develop a web-based and cloud-based platform for real-time blood pressure monitoring and analysis, including a dashboard for visualizing data and a notification system for alerting users to any abnormal readings using Spark's web development frameworks and libraries The architecture of SBP prediction systemThe architecture of SBP prediction system

9) Repeating Steps 1 to 8 of Phase 2 without updating the machine learning models and without using the ABP signal. This is done to obtain the blood pressure estimates solely from the HR, PIR, and PTT features extracted from the preprocessed data received through the wearable devices and iot sensors.

Using Apache Spark provides a scalable and efficient way to perform both the machine learning model development and the real-time data processing and web-based platform development. Spark's distributed computing capabilities allow for processing large datasets in parallel, while its built-in functions and libraries provide a range of tools for feature extraction, clustering, machine learning, signal processing, and web development.

Overall, the proposed work seeks to establish a blood pressure prediction system that operates in real-time that can be used in remote patient monitoring, personalized healthcare, and early detection of cardiovascular diseases. The proposed approach utilizes the power of Apache Spark and Apache Kafka to manage massive amounts of real-time data and provides a promising direction for improving patient outcomes and reducing healthcare costs.

A. DataSet

This study utilized the MIMIC-II dataset from Physionet, consisting of 12,000 vital signal records from US medical centers. ECG, ABP, and PPG signals are contained within the

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dataset, sampled at 125 Hz. A preprocessed version is available on Kaggle[17, 18].

B. Feature Extraction

To extract the necessary features, including PTT, PIR, HR, DBP, and SBP, from the ECG, PPG, and BP signals, the methodology used in a previous study by Farki et al. [7] will be referenced due to the volume of content involved. Fig.2 illustrates the method for extracting PTT, PIR, and HR features.

C. Clustering and Regression Models

Clustering is a common technique for grouping similar items, and the k-means is one of the famous methods in clustering. To cluster items with multiple features, the appropriate distance function must be used to measure similarity. The main purpose of clustering is to reduce the inter-item distance within each cluster. In this study, clustering was used to group independent variables extracted from signals. The determination of the optimal number of clusters was carried out by utilizing the silhouette value, following which the values of each cluster's independent variables were utilized as inputs for the regression model.

D. Random Forest Regression

Random Forest Regression is a common and user-friendly algorithm known for producing excellent results without requiring extensive parameter tuning. Random Forest is a popular ML algorithm that is frequently utilized for both classification and regression tasks. This supervised learning algorithm constructs a decision tree forest through the utilization of the bagging technique, which combines multiple models to improve the accuracy of the results. In this approach, multiple decision trees are constructed and combined to provide more accurate and consistent predictions[19].

E. Apache Kafka

Apache Kafka is a free and open-source framework for distributed streaming that makes it easier to manage the processing of massive amounts of real-time data coming from a variety of sources. It enables fault-tolerant, scalable, and efficient data processing by breaking data into smaller chunks and distributing them across a machine cluster. This project uses Kafka to handle incoming data from wearable devices and IoT sensors. The preprocessed data will be streamed to Apache Spark for real-time processing and analysis, allowing us to estimate blood pressure values using machine learning models. By combining Kafka and Spark, we can create a reliable and scalable platform for real-time blood pressure monitoring and analysis.

F. Apache Spark

Apache Spark is a framework that uses distributed computing that processes large-scale data sets in parallel. It is designed on top of Hadoop's HDFS file system and MapReduce paradigm, but provides higher-level abstractions, better performance, and more flexibility than Hadoop's MapReduce. In this project, Spark is used for both machine learning model development and real-time data processing. Specifically, Spark's MLlib library is used for machine learning model development, while Processing of real-time data is accomplished with the help of Spark Streaming. The MLlib library provides various tools, including regression, classification, clustering, and collaborative filtering algorithms. The project uses K-means clustering algorithm to cluster the extracted features of PTT, PIR, and HR, and Random Forest Regression (RFR) algorithm for each cluster's model development. Spark Streaming can be used for data processing, collecting hospital signals produced by wearable devices and IoT sensors and streaming the data through Apache Kafka. The stream processing system provided

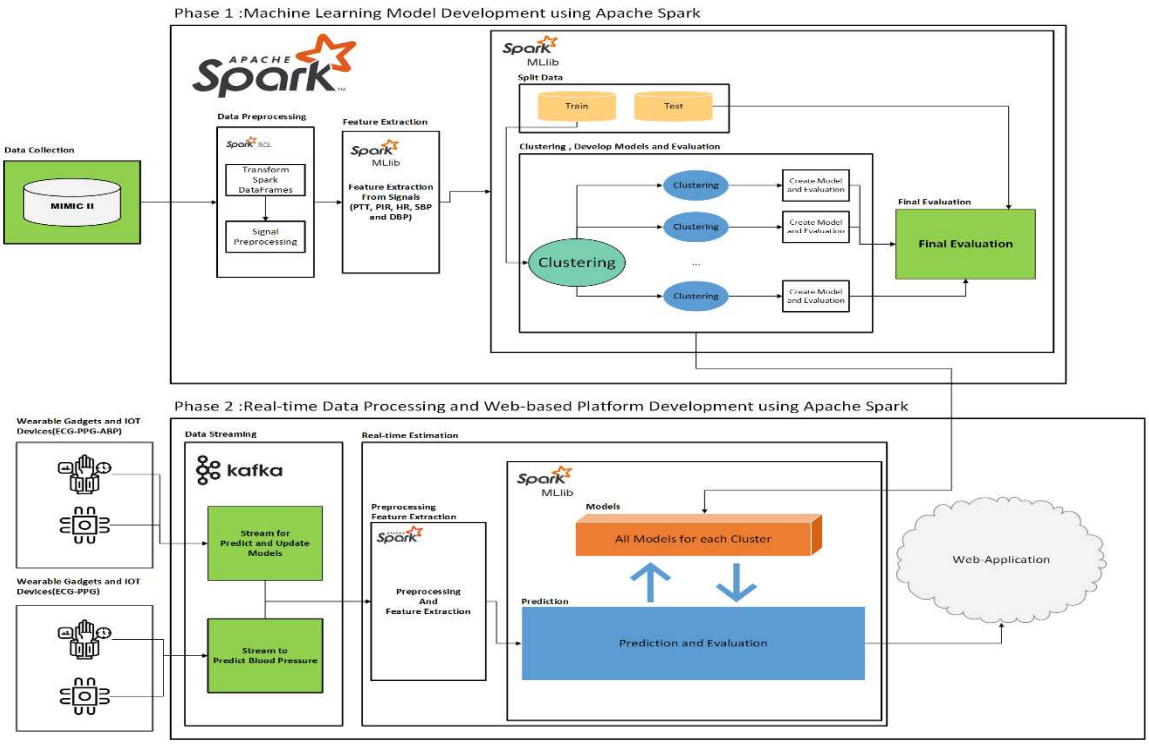

Fig. 1. The architecture of real-time blood pressure prediction system

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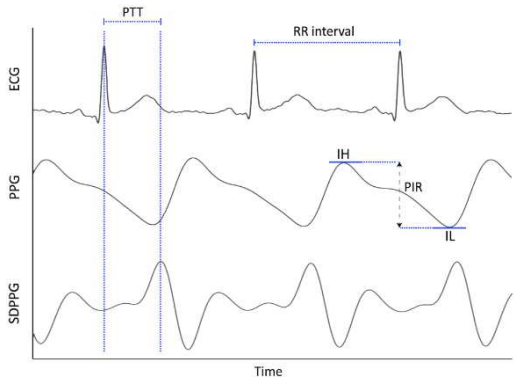

Fig. 2. The computation of the PIR and PTT ratio, as well as the HR[7].

by Spark Streaming is both robust and fault-tolerant that can handle high-volume data streams in real-time. Spark's built-in signal processing libraries are used to preprocess the signals and remove noise and artifacts. Spark's functions and libraries are used to process the data and extract features such as HR, PIR, and PTT. The machine learning models developed in the first phase estimate SBP and DBP from the extracted features. Spark's distributed computing capabilities allow for processing large datasets in parallel. Additionally, Spark's built-in web development frameworks and libraries, such as Spark SQL and Spark Streaming, are used to develop a web-based and cloud-based platform for real-time blood pressure monitoring and analysis. The platform includes a dashboard for visualizing data and a notification system for alerting users to abnormal readings.

G. Model and Results Evaluation

For this research, the performance of the developed models is evaluated using two widely recognized metrics, namely RMSE and MAE. Root Mean Square Error is a statistical measure that assesses the deviation between the predicted and actual values. It is a useful tool to evaluate model's accuracy on a given dataset. Specifically, (1) shows how to calculate RMSE[7]:

$$RMSE = \sqrt{\frac{1}{n}\sum_{j=1}^{n}(y-\hat{y})^2} \quad (1)$$

It has been widely recognized that RMSE is a robust measure for evaluating regression models in medical and bioinformatics problems due to its ability to account for the root square of the mean square error, thus reducing bias. In this study, we also employ the use of MAE as a complementary evaluation metric. Unlike RMSE, MAE is a measure of the average absolute variance between forecasted and observed values, regardless of the direction of the error. This statistical measure, also known as L1 Loss, emphasizes the magnitude of the error in estimation rather than whether it was an overestimation or underestimation. Equation (2) shows how to calculate MAE[7]:

$$MAE = \frac{\sum |y_i - \hat{y}_i|}{n} \quad (2)$$

IV. RESULTS AND DISCUSSION

In the current study, we built upon our previous research by using the same clustering approach that was previously found to significantly improve prediction results. However, we implemented this approach using Apache Spark, Spark MLlib, and Spark SQL in an offline model within the Spark environment to take advantage of the distributed and stream processing capabilities offered. In the first phase of the study, MIMIC-II data was converted into Spark data frames, and relevant features such as PTT, PIR, HR, SBP, DBP extracted from ECG, PPG, and ABP signals. These features clustered and a random forest regression model was developed for each cluster. While the results were not better than those achieved in our previous work, this was attributed to the use of less data to develop the offline model and only using a portion of MIMIC-II data for the second phase of the research. The results of the evaluation criteria for each cluster are presented in Table I. Weighted averaging was used to calculate the overall evaluation criteria.

In the second phase of the study, the lack of access to vital signs sensor data and wearable vital signs monitoring devices led us to use a portion of MIMIC-II data as simulated sensor data, comprising ECG, PPG, and ABP signals. These signals were used for real-time estimation and as a factor for updating the models. To enable real-time processing, the data were first converted into stream data using Apache Kafka and then analyzed in the Spark environment to extract features, estimate blood pressure, and update the model. Each cluster was assigned a specific Boden value, which was used to participate in the update process. To account for the limited number of samples, we divided the input samples into three parts, using these there steps to estimating blood pressure, updating the model, and evaluating the RFR model in three steps. The results of this phase can be found in Table II. Weighted averaging was used to calculate the overall evaluation criteria.

Table II shows that the model's performance improved with each update in the Spark MLlib environment. Fig.3 and Fig.4 illustrate the MAE criterion for estimating SBP and DBP for both offline and updated models 1, 2, and 3. In the subsequent step, it became possible to estimate SBP and DBP using only ECG and PPG signals. In this step, there was no need to update the model, and the results could be delivered to the user through the web-based environment.

In analyzing the literature review, the obtained results can be divided into two parts: those that developed the model without using the flow processing tool, and those that processed blood pressure based on the flow and real-time data. Our work provided better results than those of [11, 20-23] which used different deep learning models and different types of feature extraction methods and specific data for this purpose. However, in the field of works that used streaming and real-time processing methods, Saleh et al.[15] achieved better results by focusing on systolic blood pressure. This finding highlights the importance of incorporating flow and real-time data in models, which can lead to more accurate and reliable results in predicting blood pressure. These results can potentially be applied in the medical field to improve patient care and health outcomes. Further research can be conducted to explore the potential of integrating other types of data with real-time processing to enhance the performance of blood pressure prediction models. The results obtained from the current study and previous research conducted in this domain are presented in Table III.

V. CONCLUSION

In this work, we looked at how to develop an offline and online model for predicting systolic and diastolic blood pressure utilizing ECG, PPG, and ABP signals using Apache Spark, Spark MLlib, and Spark SQL. In our method, the pertinent characteristics were clustered, and a random forest regression model was created for each cluster. With a weighted average

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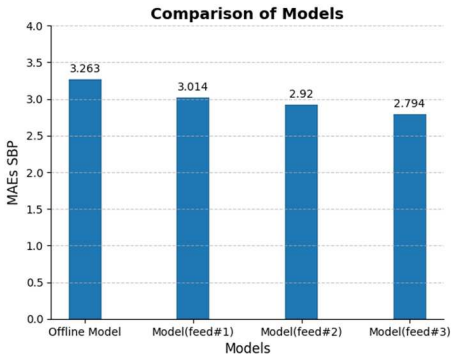

Fig. 3. MAE values of four models for the estimation of SBP

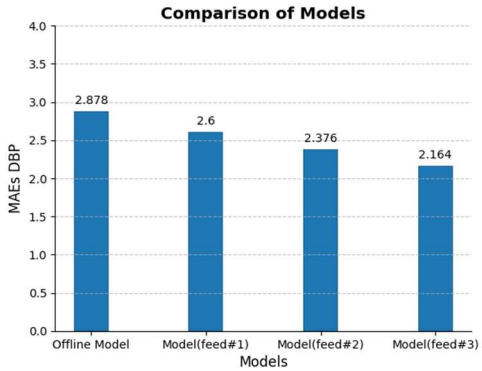

Fig. 4. MAE values of four models for the estimation of DBP

TABLE I. THE RESULTS OBTAINED FOR EACH CLUSTER AND IN TOTAL (OFFLINE)

		SBP (mmHg)			DBP (mmHg)		
Model	***Cluster***	***MAE***	***RMSE***	***r***	***MAE***	***RMSE***	***r***
RFR	Cluster1	4.257	6.867	0.84	4.034	6.114	0.88
	Cluster2	3.221	4.831	0.89	2.801	4.245	0.91
	Cluster3	3.477	5.197	0.71	2.734	5.428	0.84
	Cluster4	2.807	4.652	0.90	2.425	4.024	0.93
	Cluster5	2.009	3.891	0.86	1.751	3.901	0.89
	Total	**3.263**	**5.217**	**0.85**	**2.878**	**4.116**	**0.89**

TABLE II. THE RESULTS OBTAINED IN THE THREE STAGES OF ONLINE MODEL FEEDING

	SBP (mmHg)		DBP (mmHg)	
Feed Number	***MAE***	***RMSE***	***MAE***	***RMSE***
#1	3.014	5.201	2.6	3.502
#2	2.920	4.856	2.376	3.487
#3	2.794	4.631	2.164	3.259

MAE of 3.263 for SBP and a weighted average MAE of 2.878 for DBP, the findings from the offline model were encouraging. To update and enhance the performance of the online model, we employed simulated sensor data and real-time estimate. In the end, we were able to predict blood pressure using only ECG and PPG inputs, demonstrating that the model's performance increased with each update. Overall, The study's results demonstrate that leveraging Apache Spark, Spark MLlib, and

TABLE III. COMPARISON TO RELATED WORKS

	Method	**Machine learning comparison (DBP)**		**Machine learning comparison (SBP)**	
		MAE	RMSE	MAE	RMSE
Proposed work	Clustering & and Real-time feeding with Spark	2.164	3.259	2.794	4.631
[7]	Clustering & GBR	2.23	5.01	2.56	5.63
[20]	Multi Layer Perceptron	4.96		5.46	
[24]	ECG: hand and foot PPG: Finger	4.4	-	6.6	-
[21]	15 hidden neurons, ANN	-	-	3.03	-
[22]	Multisensor features	4.54	-	6.13	-
[23]	CNN-regression and PPG	3.45	-	5.73	-
[25]	MLR	2.82	-	2.83	-
[11]	Cnn-lstm	7.25	-	3.55	-
[10]	Cnn-lstm	2.57	-	3.44	-
[15]	Apache Spark & BI-LSTM	-	-	-	2.84

Spark SQL, along with clustering algorithms, can prove to be a viable approach for constructing both offline and online models to compute blood pressure and handle real-time data from multiple sources. This approach resulted in a noteworthy improvement in accuracy, when compared to prior techniques. Future research may explore extending the study to more varied people and use actual sensor data. It may also be advantageous to investigate the usage of additional machine learning algorithms.

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